

Cross-Encoder Concordance of Latent EEG States Across Foundation Models

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Abstract. Recent EEG foundation models learn transferable representations that support a growing range of downstream tasks; however, when applied to clinical EEG data, the organization and interpretation of their latent spaces remain poorly understood. Data-driven latent EEG states generally lack state-level clinical ground truth; therefore, states learned by different encoders cannot be assumed to represent equivalent physiological or clinical processes. In this study, we investigate whether group-associated latent-state occupancy patterns can be consistently identified across independently pretrained EEG foundation models. Using the Temple University Epilepsy Corpus (TUEP), we evaluated frozen representations from our lightweight EEG and LUNA-Base foundation model through a common pipeline comprising k -means state discovery, occupancy analysis, and occupancy-informed state alignment. At $k = 4$, the two encoders exhibited substantially different absolute state-occupancy profiles, with negligible cross-encoder correspondence for both the non-epilepsy ($r = -0.06$) and epilepsy ($r = 0.03$) groups. After one-to-one state alignment using the Hungarian assignment algorithm with cosine distance between group level occupancy signatures, the epilepsy-related occupancy change, $\Delta o = o_{\text{Epi}} - o_{\text{NonEpi}}$, showed high cross-encoder concordance across the four aligned states ($r_{\Delta} = 0.96$), with consistent directions of change across all matched state pairs. These findings demonstrate that different EEG foundation models can exhibit substantially different absolute latent-state organizations while revealing concordant group-associated occupancy patterns after alignment. Cross-encoder comparison may therefore provide an exploratory approach for evaluating clinically associated structure in learned EEG representations when state-level annotations are unavailable.

Keywords: EEG foundation models · latent-state dynamics · representation learning · cross-encoder concordance · epilepsy

1 Introduction

Electroencephalography (EEG) is a common non-invasive technique for measuring and monitoring the electrical activity of the brain. It has been extensively

applied to neurological and physiological analysis, including epilepsy and seizure detection, sleep monitoring, stress assessment, and emotion recognition [14, 5, 10, 2]. However, the low signal-to-noise ratio, non-stationarity, and substantial inter-subject variability of EEG signals continue to challenge the development of models that generalize across subjects, datasets, and downstream tasks [9, 7]. Recent EEG foundation models have emerged as a promising approach to this problem by learning transferable representations from large-scale EEG data. Instead of training a task-specific model directly from raw EEG, some foundation models transform EEG signals into learned latent representation spaces that can be adapted to multiple downstream applications [15, 12, 3, 4]. Such representation learning aims to reduce the dependence on task-specific feature engineering and improve model transferability across EEG domains. Although increasing attention has been devoted to improving downstream predictive performance, the clinical interpretability of these learned representations remains considerably less understood [7]. In particular, when latent representations contribute to improved prediction, it remains unclear what structures within the latent space represent physiologically and whether the discovered patterns are reproducible across different EEG foundation models [11, 8].

This creates a central challenge for the clinical analysis of learned EEG representations: latent states discovered within these representation spaces generally lack direct state-level clinical ground truth. Clinical EEG datasets may provide subject-level labels, such as epilepsy versus non-epilepsy, but they typically do not specify the physiological meaning of individual latent states. Therefore, improved predictive performance alone does not establish that the learned latent states correspond to clinically meaningful neural patterns. Therefore, the learned state cannot be assumed to represent a particular neurological condition or physiological process without additional validation. Furthermore, because independently trained encoders may learn different representation spaces, the identity and organization of their latent states can be encoder-dependent. This raises an important question for the use of EEG foundation models beyond black-box prediction: whether disease-related patterns discovered in their latent representations are reproducible across independently learned representation spaces.

To address this question, we first develop a lightweight EEG foundation model designed to learn transferable representations from large-scale unlabeled EEG recordings. The proposed model is pretrained on the Temple University Hospital EEG Corpus (TUEG) and subsequently evaluated under a frozen-encoder protocol. Beyond downstream classification performance, we investigate whether the latent representations learned by our model capture clinically associated structure that is reproducible across independently pretrained EEG encoders. Specifically, we compare our model with LUNA-Base [3], an established EEG foundation model, by evaluating epilepsy-related changes in latent-state occupancy on the Temple University Epilepsy Corpus (TUEP). Both encoders are frozen and evaluated on TUEP, enabling a direct comparison of their learned latent representations without task-specific adaptation. Rather than assigning clinical meaning directly to individual latent states, we compare relative

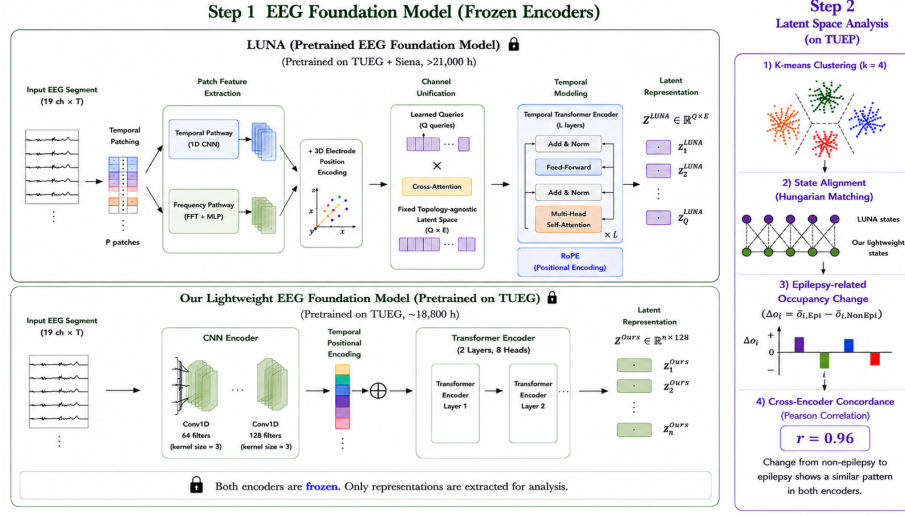


Fig. 1. Overview of the proposed cross-encoder latent-state analysis framework. TUEP EEG recordings are preprocessed and independently encoded using two frozen pre-trained EEG foundation models, LUNA and our lightweight model. The resulting latent representations are clustered using k -means ($k = 4$), followed by cross-encoder state alignment. Epilepsy-related occupancy changes are quantified as $\Delta o_i = \bar{o}_{i,\text{Epi}} - \bar{o}_{i,\text{NonEpi}}$ and compared across the aligned representation spaces. The two encoders exhibit high concordance in the epilepsy-related occupancy-change pattern ($r = 0.96$).

group-associated changes between epilepsy and non-epilepsy after aligning the independently learned state spaces. Specifically, for each encoder, we define the epilepsy-related occupancy change of latent state i as $\Delta o_i = \bar{o}_{i,\text{Epi}} - \bar{o}_{i,\text{NonEpi}}$. We then evaluate the cross-encoder concordance of these occupancy-change patterns after state alignment. This formulation allows us to examine whether independently learned representation spaces capture consistent epilepsy-related changes without assuming that their individual latent states possess predefined clinical or physiological meanings.

The main contributions of this study are threefold: (1) Model: we develop a lightweight EEG foundation model for transferable representation learning and evaluate its latent representations against LUNA-Base, an independently pretrained EEG foundation model, under a common frozen-encoder protocol. (2) Method: we formulate cross-encoder concordance as an evaluation strategy for latent EEG representations when direct state-level clinical annotations are unavailable, using an occupancy-informed framework to align independently learned latent states and quantify epilepsy-related changes in state occupancy. (3) Finding: on TUEP, we show that the two encoders exhibit substantially different absolute latent-state occupancy profiles, with negligible cross-encoder correspondence for both non-epilepsy ($r = -0.06$) and epilepsy ($r = 0.03$). Fol-

lowing state alignment, the epilepsy-related occupancy-change patterns exhibit high cross-encoder concordance ($r_{\Delta} = 0.96$), with consistent directions of change across all four matched state pairs. These results demonstrate the potential of occupancy-informed alignment for evaluating concordant group-associated patterns across independently learned EEG representation spaces despite differences in their absolute latent-state organization.

2 Methods

2.1 EEG Foundation Model Encoders

To investigate whether epilepsy-related latent-state patterns are reproducible across independently learned EEG representation spaces, we evaluated two pretrained EEG foundation model encoders: our lightweight universal semantic EEG foundation model and LUNA-Base. The two models differ substantially in architecture and model size, providing distinct representation spaces for cross-encoder comparison.

Our lightweight EEG foundation model adopts a CNN-Transformer architecture adapted from the universal semantic feature extraction framework proposed by Ahmadi and Mesin [1]. The encoder consists of two one-dimensional convolutional layers followed by temporal positional encoding and a two-layer Transformer encoder. In this study, we extend this architecture to large-scale self-supervised EEG representation learning by pretraining the model on approximately 18,800 hours of EEG recordings from the TUEG using a signal-reconstruction objective. The reconstruction loss minimizes the difference between the input EEG segment and its reconstructed signal. The resulting encoder contains 1.46 million parameters and produces a 128-dimensional latent representation for each EEG segment. For cross-encoder comparison, LUNA-Base was selected as an independently pretrained EEG foundation model. In contrast to our lightweight CNN-Transformer architecture, LUNA employs patch-level temporal and frequency feature extraction, channel unification, and Transformer-based temporal modeling to construct its latent representations. LUNA-Base contains 7.10 million parameters and produces a 256-dimensional latent representation for each EEG segment. Table 1 summarizes the two encoders evaluated in this study.

Table 1. Comparison of the evaluated EEG foundation model encoders.

Model	Parameters	Backbone	Latent Dim.
Our lightweight model	1.46M	CNN-Transformer	128
LUNA-Base	7.10M	LUNA encoder	256

For the present study, both pretrained encoders were frozen and used only for representation extraction. Given an EEG segment, the two encoders independently generated latent representations. No model parameters were updated

using TUEP data or epilepsy labels. This frozen-encoder design allows us to examine whether group-related patterns emerge consistently across independently learned representation spaces without introducing task-specific adaptation.

2.2 Dataset and Study Design

Two EEG datasets were used in this study for our model pretraining and latent-state evaluation, respectively: TUEG and TUEP. Following previous EEG foundation model studies [16, 6, 13], EEG recordings were band-pass filtered between 0.3 and 75 Hz, notch filtered at 60 Hz, mapped to the standard 19-channel international 10–20 montage, resampled to 200 Hz, and segmented into non-overlapping 30-second windows. Signal amplitudes were normalized to approximately $\pm 100 \mu\text{V}$ and subsequently scaled to the range $[-1, 1]$.

TUEG was used for self-supervised pretraining of our lightweight EEG encoder. The pretraining dataset comprised 2,259,998 EEG segments, corresponding to approximately 18,800 hours of recordings. The model was implemented using Python 3.12.12 and PyTorch 2.5.1 with CUDA 12.4 and trained on four NVIDIA A100 GPUs using Distributed Data Parallel (DDP). Pretraining used the Adam optimizer with an initial learning rate of 1×10^{-4} , a batch size of 64. The reconstruction loss was defined using the mean squared error (MSE) between the original EEG segment $\mathbf{x}$ and the reconstructed signal $\hat{\mathbf{x}}$ as $\mathcal{L}_{\text{MSE}} = \frac{1}{T} \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2$, where T denotes the number of temporal samples. Optimizing this reconstruction objective encourages the encoder to learn latent representations that preserve information necessary to reconstruct the input EEG signal. Training was performed for up to 100 epochs with early stopping (patience = 5), requiring approximately 100 hours in total. TUEP was subsequently used for latent-state analysis (Fig.1). Recordings were divided into epilepsy and non-epilepsy groups, with $n = 100$ subjects in each group. For downstream classification, five-fold cross-validation was performed, with 80% of the subjects used for training and 20% for validation in each fold, while the pretrained encoder remained frozen throughout the evaluation. EEG recordings were processed into fixed-length segments using a common channel configuration and preprocessing protocol compatible with both encoders. The same TUEP EEG segments were independently passed through the frozen LUNA-Base and our lightweight EEG encoders. No fine-tuning or task-specific adaptation was performed on TUEP.

The resulting latent representations were analyzed using the same downstream clustering and state-analysis pipeline, allowing cross-encoder differences to reflect the independently learned representation spaces rather than task-specific adaptation. Clinical labels were used only to define the epilepsy and non-epilepsy groups. Because no state-level clinical annotations were available, individual latent states were treated as data-driven representation states and were not assigned predefined diagnostic or physiological meanings. Instead, we evaluated whether the relative change in latent-state occupancy from non-epilepsy to epilepsy was reproducible across the two independently learned representation spaces.

2.3 Latent-State Discovery, Alignment, and Occupancy Analysis

For each encoder $e \in \{\text{LUNA}, \text{Ours}\}$, frozen segment-level embeddings were independently partitioned into k latent states using k -means clustering. The number of clusters was varied from $k = 3$ to 20, and the within-cluster sum of squares (WCSS) was computed separately for LUNA-Base and our lightweight model. Based on the elbow observed in the WCSS curve, $k=4$ was selected for the primary analysis. Let

$$z_{s,t}^{(e)} \in \{0, \dots, k-1\}$$

denote the latent-state assignment of segment t from subject s under encoder e .

For each subject, the occupancy of latent state i was defined as the proportion of EEG segments assigned to that state:

$$o_{s,i}^{(e)} = \frac{1}{T_s} \sum_{t=1}^{T_s} \mathbb{I}[z_{s,t}^{(e)} = i],$$

where T_s denotes the total number of EEG segments from subject s . Group-level mean occupancy $\bar{o}_{i,G}^{(e)}$ was then calculated separately for the epilepsy and non-epilepsy groups.

Because k -means clustering was performed independently within each representation space, the resulting state labels are arbitrary. For example, State 1 identified by LUNA cannot be assumed to correspond directly to State 1 identified by our lightweight model. We therefore performed occupancy-informed state alignment before comparing group-related patterns across encoders.

For each latent state i , we defined its group-level occupancy signature as

$$f_i^{(e)} = \left(\bar{o}_{i,\text{NonEpi}}^{(e)}, \bar{o}_{i,\text{Epi}}^{(e)} \right),$$

which characterizes how frequently that state occurred in the non-epilepsy and epilepsy groups. A cross-encoder cost matrix was constructed using cosine distance between the occupancy signatures:

$$C_{ij} = 1 - \cos \left(f_i^{(\text{LUNA})}, f_j^{(\text{Ours})} \right).$$

The Hungarian assignment algorithm was then used to identify the one-to-one state correspondence that minimized the total cross-encoder cost:

$$\pi^* = \arg \min_{\pi} \sum_{i=0}^{k-1} C_{i,\pi(i)}.$$

Thus, each LUNA state was matched to exactly one state from our lightweight model, with each state used only once. The resulting optimal permutation π^* was used to reorder the states from our lightweight model according to the corresponding LUNA state order.

To characterize group-related changes rather than absolute state occupancy, the epilepsy-related occupancy change for each state was defined as

$$\Delta o_i^{(e)} = \bar{o}_{i,\text{Epi}}^{(e)} - \bar{o}_{i,\text{NonEpi}}^{(e)}.$$

Positive values indicate increased state occupancy in epilepsy, whereas negative values indicate reduced occupancy relative to the non-epilepsy group.

2.4 Cross-Encoder Concordance and Evaluation Metrics

After applying the optimal state permutation, the aligned occupancy-change vector from our lightweight model was defined as

$$\Delta \tilde{o}_i^{(\text{Ours})} = \Delta o_{\pi^*(i)}^{(\text{Ours})}.$$

Cross-encoder concordance of the epilepsy-related occupancy changes was then quantified using Pearson correlation:

$$r_\Delta = \text{corr} \left(\Delta \mathbf{o}^{(\text{LUNA})}, \Delta \tilde{\mathbf{o}}^{(\text{Ours})} \right).$$

Because π^* was estimated from the same group-level occupancy signatures used to define Δo , r_Δ represents concordance following occupancy-informed alignment rather than an independent validation of clinical-state correspondence.

To examine the organization of the learned representation spaces, latent embeddings from both encoders were visualized using principal component analysis (PCA), retaining 95% of the cumulative variance, and t-distributed stochastic neighbor embedding (t-SNE), with EEG segments colored according to their k -means state assignments. Frozen-encoder downstream classification performance was additionally evaluated on TUAB and TUEP using five-fold cross-validation. Performance was quantified using area under the receiver operating specificity (SPE), accuracy (ACC), sensitivity (SEN), and F1-score. No encoder fine-tuning was performed during downstream evaluation.

3 Results

3.1 Frozen-Encoder Classification Performance

We first evaluated the classification performance of the two frozen encoders on TUEP. As shown in Table 2, the two encoders achieved comparable overall performance but exhibited different sensitivity–specificity profiles. Our lightweight model achieved a SPE of 63.00 ± 17.20 , ACC of 69.00 ± 3.24 , SEN of 75.00 ± 14.14 , and F1-score of 70.37 ± 4.01 . In comparison, LUNA-Base achieved a SPE of 73.00 ± 10.30 , ACC of 67.50 ± 2.74 , SEN of 62.00 ± 7.48 , and F1-score of 65.46 ± 2.89 . Our lightweight model therefore showed higher sensitivity, accuracy, and F1-score, whereas LUNA-Base achieved higher specificity. These results indicate that our lightweight model maintains competitive frozen-transfer performance despite its substantially smaller parameter count.

Table 2. Frozen-encoder classification performance on TUEP (mean $\pm$ std). The best result for each metric is highlighted in **blue bold**.

Metric	Our Lightweight Model	LUNA-Base
SPE	63.00 $\pm$ 17.20	73.00 $\pm$ 10.30
ACC	69.00 $\pm$ 3.24	67.50 $\pm$ 2.74
SEN	75.00 $\pm$ 14.14	62.00 $\pm$ 7.48
F1	70.37 $\pm$ 4.01	65.46 $\pm$ 2.89

3.2 Latent Representation and State Visualization

The latent representations learned by the two models were visualized using PCA and t-SNE at $k = 4$, as shown in Figure 2. Both encoders exhibited structured organization of EEG segments within their respective representation spaces, while the geometry and distribution of the four latent states differed between models. These observations indicate that the two independently pretrained encoders organize their latent representations differently, motivating the need for explicit state alignment before cross-encoder comparison. Although these visualizations illustrate the organization of the learned latent spaces, they should not be interpreted as evidence that individual latent states possess predefined clinical or physiological meanings.

3.3 State Occupancy Across Clinical Groups

The latent-state occupancy distributions at $k = 4$ for the epilepsy and non-epilepsy groups are summarized in Figure 3. The two encoders exhibited substantially different absolute occupancy profiles, indicating that LUNA-Base and our lightweight model organize their latent representation spaces differently. Absolute state occupancy showed negligible cross-encoder correspondence for both the non-epilepsy ($r = -0.06$) and epilepsy ($r = 0.03$) groups. These differences motivate state alignment before evaluating group-associated occupancy patterns across the two representation spaces. Within each representation space, LUNA-Base showed significant differences between the epilepsy and non-epilepsy groups in State 0 (FDR-adjusted $p = 0.033$) and State 2 (FDR-adjusted $p = 0.032$), whereas state-wise occupancy differences from our lightweight model did not remain significant after FDR correction. These findings indicate that individual state-occupancy patterns depend on the learned representation space and that states with the same numerical labels should not be assumed to represent equivalent latent patterns across encoders.

3.4 Occupancy-Informed State Alignment and Cross-Encoder Concordance

Because the latent states were independently learned by the two encoders, we next aligned the states using cosine distance between their group-level occupancy

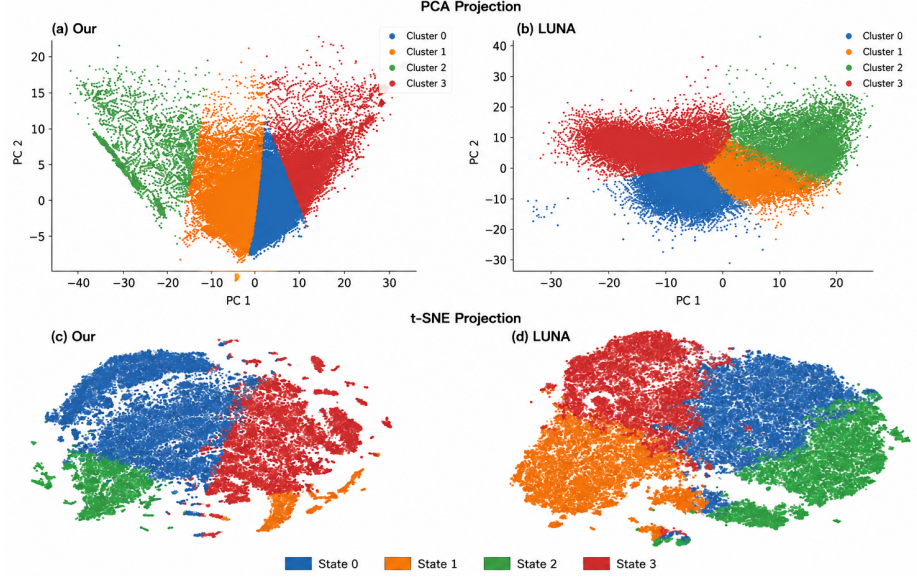


Fig. 2. Latent-state representations learned by our lightweight model and LUNA-Base on TUEP at $k = 4$. (a) PCA visualization of our lightweight model; (b) PCA visualization of LUNA-Base; (c) t-SNE visualization of our lightweight model; and (d) t-SNE visualization of LUNA-Base. Colors indicate the four latent states identified by k -means clustering.

signatures. The resulting one-to-one correspondence is summarized in Table 3. LUNA States S0–S3 were matched to States S3, S0, S1, and S2, respectively, from our lightweight model.

We then examined the relative occupancy change from non-epilepsy to epilepsy, defined as

$$\Delta o_i = \bar{o}_{i,\text{Epi}} - \bar{o}_{i,\text{NonEpi}}.$$

As shown in Fig. 4(a), the direction of occupancy-change was consistent across all four aligned state pairs. Aligned State S1 showed the largest decrease in occupancy from non-epilepsy to epilepsy in both encoders, whereas aligned States S0, S2, and S3 showed positive occupancy changes.

Examining the cross-encoder correlation in greater detail, the aligned occupancy-change vectors exhibited high concordance across the four aligned states ($r_\Delta = 0.96$), as illustrated in Fig. 4(b). Thus, although the two encoders exhibited different absolute state-occupancy profiles, occupancy-informed alignment revealed a similar relative epilepsy–non-epilepsy occupancy pattern across their respective latent representation spaces. The state permutation was estimated using the same group-level occupancy signatures that contribute to the calculation of Δo . Therefore, $r_\Delta = 0.96$ can be interpreted as high concordance following occupancy-informed alignment, suggesting that clinically associated patterns

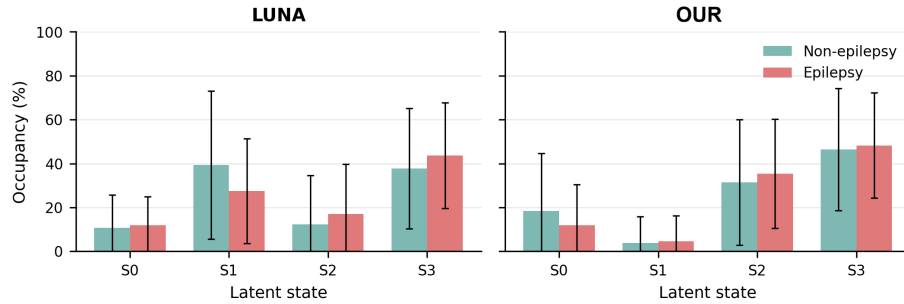


Fig. 3. Latent-state occupancy distributions for the non-epilepsy and epilepsy groups at $k = 4$ using LUNA-Base (left) and our lightweight model (right). Bars represent group-level mean occupancy, and error bars indicate variability across subjects.

Table 3. Hungarian state matching using cosine distance on group-level occupancy signatures. Δo denotes the occupancy change from non-epilepsy to epilepsy.

LUNA State	Matched Our State	LUNA Δo (%)	Our Δo (%)	Cosine Distance
S0	S3	+1.30	+1.85	0.0007
S1	S0	−11.88	−6.61	0.0008
S2	S1	+4.63	+0.76	0.0022
S3	S2	+5.96	+4.00	0.0001

may be consistently identified across the two latent representation spaces and providing a potential step toward the clinical interpretation of EEG foundation-model representations.

4 Discussion

This study investigated whether independently pretrained EEG foundation models exhibit concordant group-associated patterns within their latent representations. Although LUNA-Base and our lightweight model achieved comparable frozen-encoder classification performance on TUEP, their latent spaces showed substantially different state organizations and absolute occupancy profiles. These findings highlight an important challenge for the interpretation of EEG foundation models: latent states learned by different encoders should not be assumed to have directly corresponding identities, even when the models achieve similar downstream performance.

Despite these differences, occupancy-informed state alignment revealed a similar pattern of epilepsy-related occupancy change across the two representation spaces. After one-to-one state matching, all four aligned state pairs exhibited consistent directions of occupancy change from non-epilepsy to epilepsy.

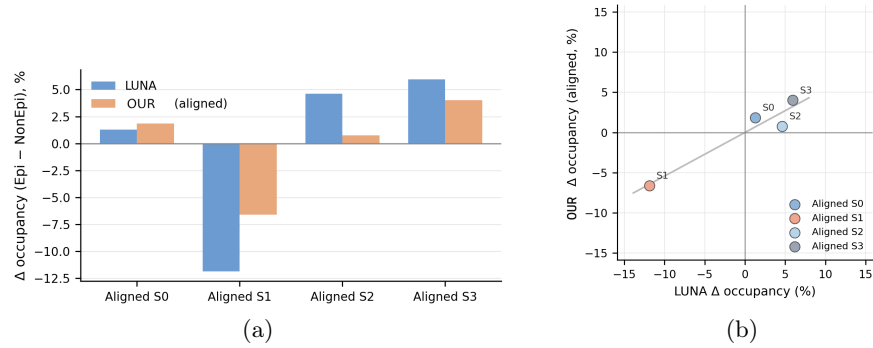


Fig. 4. Cross-encoder comparison of epilepsy-related latent-state occupancy changes after occupancy-informed state alignment. (a) Aligned occupancy changes ($\Delta o = \text{Epi} - \text{NonEpi}$) for LUNA-Base and our lightweight model across the four matched latent states. (b) Relationship between the aligned occupancy-change vectors, showing high concordance ($r_{\Delta} = 0.96$).

The aligned occupancy-change vectors showed high cross-encoder concordance ($r_{\Delta} = 0.96$). In particular, the aligned State S1 showed the largest reduction in occupancy in both models, whereas the remaining aligned states showed positive occupancy changes. These observations suggest that relative group-associated changes may provide a useful level of comparison between independently learned EEG representation spaces, even when their absolute state organizations differ. From a clinical perspective, this framework provides an exploratory approach for examining latent representations when state-level clinical annotations are unavailable. EEG foundation models typically project recordings into high-dimensional representation spaces optimized for transferable feature learning, but the physiological meaning of individual latent dimensions or data-driven states is generally unknown. Rather than assigning a diagnostic meaning to individual states, the present analysis evaluates how the occurrence of these states changes between clinically defined groups.

5 Conclusion

Although LUNA-Base and our lightweight EEG foundation model exhibited substantially different absolute latent-state occupancy profiles, occupancy-informed state alignment revealed highly concordant epilepsy-related occupancy-change patterns across the two representation spaces ($r_{\Delta} = 0.96$). These findings suggest that relative disease-associated patterns may provide a useful basis for comparing independently learned EEG representation spaces, even when their absolute latent-state organization differs. This group-associated analysis may provide an intermediate step between foundation-model representations and clinically interpretable EEG analysis.

6 Limitation

Several limitations should be considered. First, the state correspondence was determined using group-level epilepsy and non-epilepsy occupancy signatures. Because the same occupancy information contributes to both the alignment and the subsequent calculation of Δo , the observed $r_{\Delta} = 0.96$ is not an independent validation of cross-model state equivalence. Second, the correlation is calculated across only four aligned latent states at $k = 4$ and should therefore be interpreted as an exploratory descriptive measure rather than strong statistical evidence of generalizable cross-encoder reproducibility. Third, the present analysis uses recording-level clinical groups without state-level physiological annotations, preventing direct interpretation of the discovered states in terms of seizure activity or specific electrophysiological mechanisms.

Future work will focus on evaluation whether the observed group-associated patterns remain consistent under alignment criteria independent of clinical group occupancy, across different choices of k , and in independent EEG datasets. Incorporating temporally resolved seizure annotations and established EEG biomarkers could further determine whether particular latent-state changes correspond to physiologically meaningful neural activity. Extending the analysis to additional EEG foundation models would also help determine whether cross-encoder concordance reflects a broader property of learned EEG representations rather than agreement between two specific models.

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